

4. The UK government aims to remove the sale of petrol and diesel cars by 2030. It is expected that by 2027 more electric cars will be sold than any other type.

- (a) The data in **Table 1** shows how the power generated in a typical petrol engine is converted.

Table 1. Efficiency of a petrol driven car

	Description of energy conversion	% Power
engine	heat loss	70
movement	noise	1
	friction	6
	power to the wheels	
	total power	100

- (i) Use the information in **Table 1** to calculate the % power to the wheels. [1]

% power to the wheels =

- (ii) A petrol car generates 66 kW of power.

Use your answer to (a)(i) to calculate the actual power delivered to the wheels. [2]

actual power to the wheels = kW



- (iii) State **two** ways in which engineers have improved the design of cars to improve their efficiency. [2]

1.

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2.

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- (b) One of the biggest problems with electric vehicles is the distance travelled per charge.

Table 2 gives information about three types of battery chargers in use.

Table 2

Type of charger	Charger 1	Charger 2	Charger 3
operating voltage (V)	V_1	240	480
operating current (A)	20	80	I_3
power delivery (W)	2400	19200	144 000
distance travelled per hour of charging (km)	3	83	545

- (i) Use the equation

$$\text{voltage} = \frac{\text{power}}{\text{current}}$$

to calculate the operating voltage, V_1 , of charger 1.

[2]

$$V_1 = \dots\dots\dots \text{ V}$$



(ii) Use the equation

$$\text{current} = \frac{\text{power}}{\text{voltage}}$$

to calculate the current, I_3 , of charger 3.

[2]

$I_3 = \dots\dots\dots$ A

(iii) Electric car companies claim that it is cheaper to run their electric cars than petrol ones.

Use the following information to explain if this claim is correct.

[3]

- A typical petrol car needs 50 litres of fuel to travel 545 km. The cost of one litre of petrol is £1.30.
- An electric car requires 144 kWh of electricity to travel 545 km. The cost of one unit of electricity is £0.30.
- cost of electricity (£) = cost per unit (£) $\times$ number of units (kWh)

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